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Simio Project
Simio Shelving Shop Problem Formulation and Analysis



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Introduction

The Simio Shelving Shop is currently experiencing low order fill rates, affecting the shops assembly process and bottom line results. A dynamic buffering and inventory paradigm system, Demand Driven Material Requirement Planning (DDMRP), was recommended by an outside consultant to address the rampant stockout issue.

The shop currently uses a static buffer level that results in frequent stockouts of finished goods and supplies. The Dynamic Buffering System will improve Key Performance Indicators (KPIs). DDMRP can become a better method of monitoring risks in the buffer levels.

Data from Sales, Quality, Manufacturing, and suppliers will be used to determine the buffers location and modify the parameters to improve KPIs. To prevent future stockouts, the prediction of the KPIs change after integration of the buffers in the factory process. For an improvement in the raw materials sector, suppliers will be reevaluated through their varying lead times and quality.

Information Record

Study Reasons

1. Virtual simulation of the Simio Shelving Shop assembly process and possible improvements.
2. Analyze assembly Throughput and possible risks in the process.
3. Determine an average unit time in the system.
4. Determine the Workstation utilization and its efficiency to the process.
5. Establish an average inventory cost for each buffer.
6. Calculate a number of stockouts.

Production Characteristics

Sales Analysis (third quarter 2020):

To analyze the production performance the sales results can be analyzed to determine what product was produced the most and what months required better throughput.

Table 1. Quantity of shelves ordered in 2020.

Product Type	**Quantity Ordered**
B22	1450
B23	1457
B24	1161
O22	1226
O23	1286
O24	990
Grand Total	**7570**

Chart 1. Quantity Ordered by Product in 2020



Table 2. Quantity Ordered by month in 2020

Month	Quantity Ordered
Sep	**376**
Oct	**2761**
Nov	**2688**
Dec	**1745**

Chart 2. Quantity Ordered by month in 2020

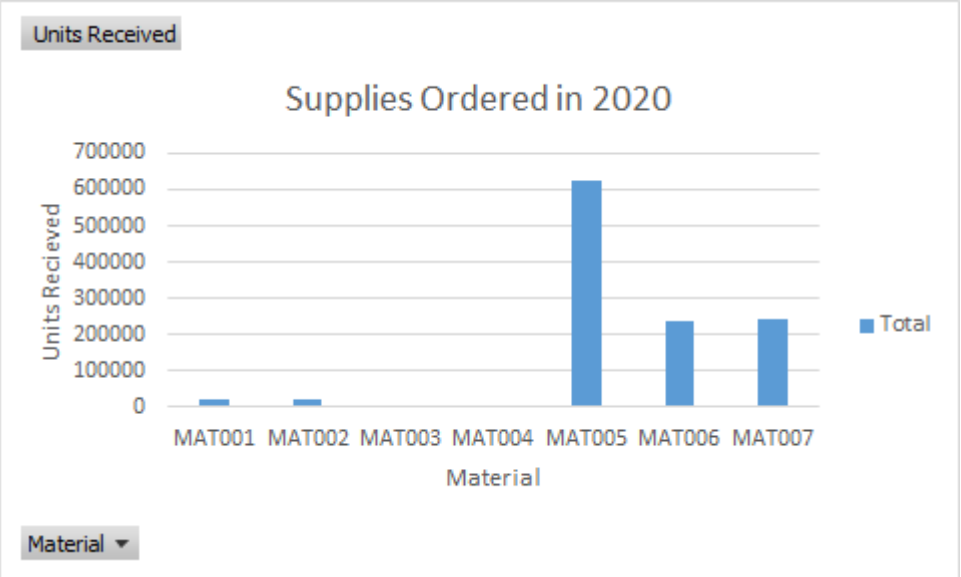


Supplier Data Analysis (third quarter 2020)

Table 3. Supplies Ordered in 2020

Material	Units Received
MAT001	23983
MAT002	20920
MAT003	764
MAT004	666
MAT005	624388
MAT006	236500
MAT007	243912
Grand Total	1151133

Chart 3.Supplies Ordered in 2020



Key Performance Indicators.

The KPIs that the managers are interested in are:

- average inventory cost
- order fill rate
- throughput
- average unit production time

For a better comprehension of each KPIs and a better analysis of its function, they can be rearranged by their objective and type. Effectivity and Efficiency are the two objectives that the process has to maximize its utility. Depending on the type the KPIS can be analyzed in a different way to determine the best way to improve it.

Table 4. KPIs Classification

Indicator	Objective	Type
Average inventory cost	Effectivity	Reason
Order fill rate	Effectivity	Reason
Throughput	Efficiency	Reason
Average unit production time	Efficiency	Direct Data

Work sampling (Direct observation).

Blackbox analysis

To have a basic perspective of the assembly process, a balck box diagram is introduced to determine where direct observation is needed.

Figure 1 Blackbox analysis



Environment Identification

- The different raw materials that each supplier offers us which will affect the quality and price of the final product
- Production Characteristics:
  - Divided in 3 subassemblies for the back panel, frame and shelf, all then go to a final assembly, all of this can be seen more clearly in **Figures 2 and 3**
- Human Factor Characteristics:
  - The personnel consists of 33 employees, one per station. Every one of them is trained for multiple tasks, so they are assigned to a different one daily in order to reduce the repetitions. They must go pick up the material to the inventory to perform a task, and then deliver the product to the next station. Work hours are from 9am to 5pm with lunchtime from 12pm to 1pm.

System Resources Identification

- **Part Design**

- Analyze the materials of the suppliers to determine which is the one that offers the best quality of wood.
- Some designs are more complicated to manufacture as they depend on the shelf specifications

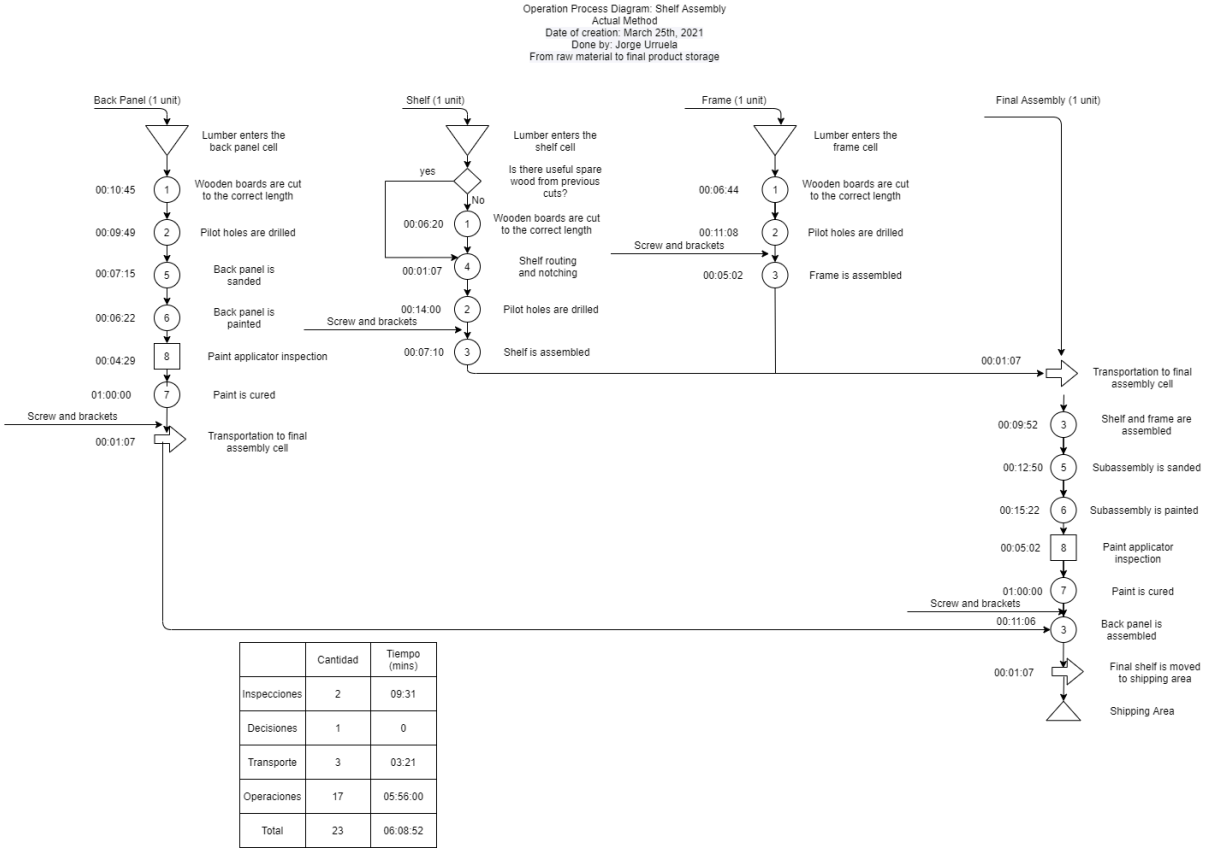
- **Material**

Table 1: Unit cost and material description for each current supplier

Supplier	Cost per Unit ($)	Material Description
Incity Timber	6,24	Birch 3/4" x 14" x 54"
Highe´s Lumber	7,61	Oak 3/4" x 14" x 54"
Local Storage Hardware	23,00	Birch Paint (gal)
Jone´s Paint Store	27,00	Oak Paint (gal)
Building Center	0,04	#8 x 1/2" wood screws
King´s Supply	0,07	#12 x 2-1/2" wood screws
Club Hardware	0,52	Brackets

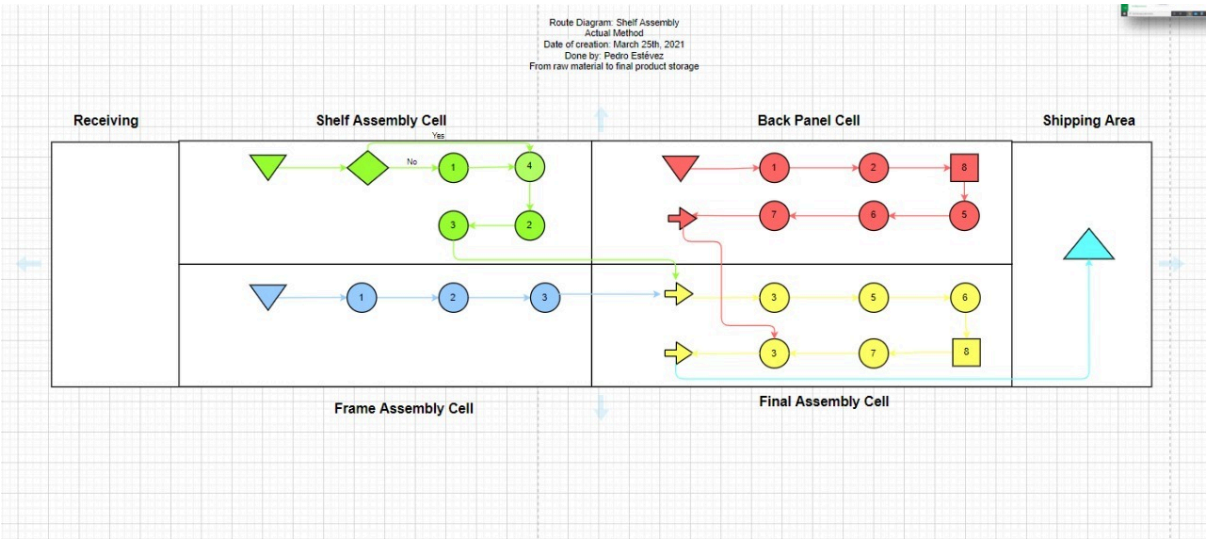
- **Manufacturing Sequence and Process**

Figure 2 Operation Process and Flow Process diagram



	Cantidad	Tiempo (mins)
Inspecciones	2	09:31
Decisiones	1	0
Transporte	3	03:21
Operaciones	17	05:56:00
Total	23	06:08:52

Figure 3 Stroke diagram



- ❖ **Frame Assembly:**

-Processing time of the cutting and drilling process for each frame varies by shelf configuration.

-The frame assembly time does not vary by shelf configuration.

- ❖ **Shelf Assembly:**

-Processing times vary by shelf configuration.

-The processing time to complete the routing for each assembly varies by shelf configuration and can be found through analysis of the data.

-When scrap wood is used in this way, the number of cuts is decreased, which decreases the overall processing time

- ❖ **Back Pannel:**

-The time to cut, drill, sand, and paint back panels varies by the shelf configuration, and will need to be collected through investigation of the data file attachment.

-The processing times for the assembly, sanding, painting, and attachment of the back panel vary by the shelf configuration.

- **Settings and Tools**

When wood needs to be cut, the worker needs to take a moment to dust off the machine, secure the wood, and adjust the cutting instruments to the correct measurements each time the cutting process is started for an assembly.

The processing time to complete the routing for each assembly varies by shelf configuration.

The wooden boards must be cut to the correct length of the shelf designPilot holes drilled for assembly

machinery and equipment Analysis

Table 5. Analysis of the machinery and equipment used in the Shelf assembly process.

Equipment and machinery	**Use**	**Number of stations that use it**
drills	Assembly the shelf together.	3
screws	Keep the shelves together.	3
paint	Give the shelf a color of presentation	1
cutting instrument	cut the woods to their correct length	4

The frame is assembled using screws and and brackets

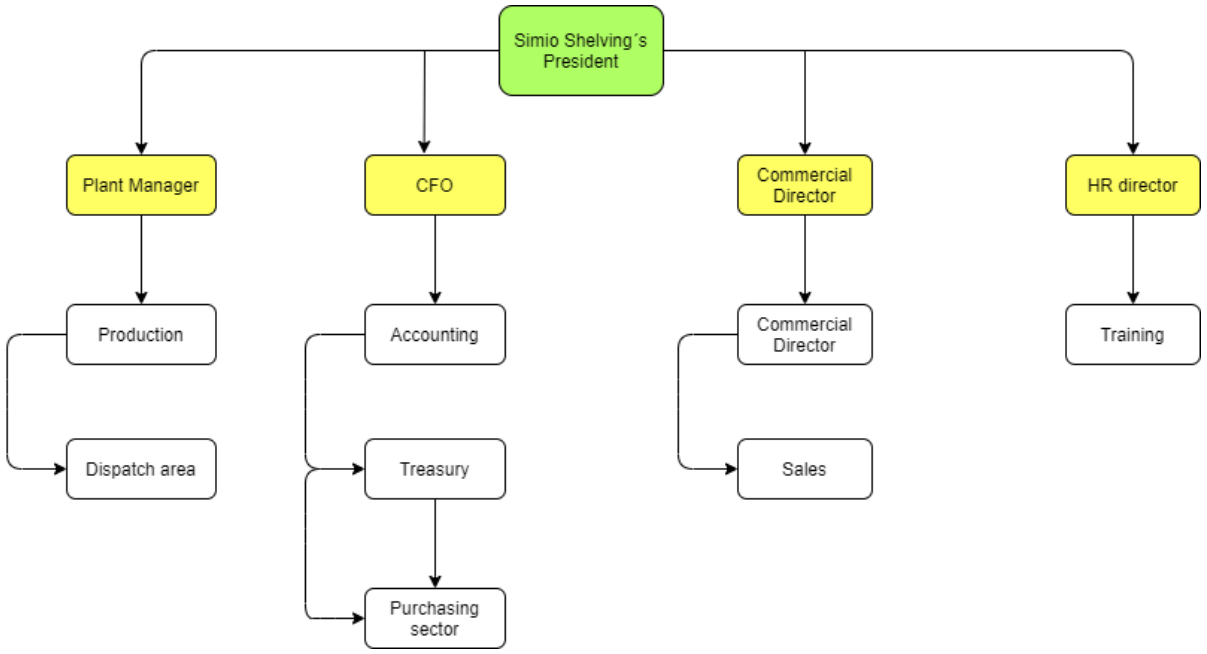
- **Plant Distribution**

The process path diagram could be analyzed to identify subassemblies or bottlenecks that are having delays and take advantage of those delays to conduct inspections.

Identifying components or functions of the system

- ❖ Demand is the most important component of the system. Since it determines the amount of shelf to be manufactured

System administration identification



- Decisions are made by the managers, after General Manager's approval

Object Implementation

Revamp the facility's buffering system by incorporating a DDMRP dynamic inventory system for each buffer. Integrate a buffer configuration that will maximize the shop's order fill rate and minimize the shop's average inventory cost. Create or remove buffers, provide all necessary stations access to the required material by DDMRP dynamic replenishment policy. For the dynamic inventory system.

Metrics To Analyze

• Throughput
• Average unit time in system
• Workstation utilization
• Average inventory cost for each buffer
• Number of stockouts
• Unmet demand by day

Data Analysis

- **Results for each Supplier**

Table 6: Summarized Supplier Data during the trial period with Pacifica Timber

Supplier	Units Received	% of inspected units
Building Center	263557	84,49%
Club Hardware	92920	80,71%
Incity Timber	13212	82,77%
Jone's Paint Store	174	100,00%
King's Supply Company	84255	92,21%
Local Storage Hardware	284	100,00%
Pacifica Timber	7719	83,07%
Totals	**462121**	**85,08%**

Table 7: Summarized Quality Data for each supplier during the trial period with Pacifica Timber

Supplier	Units Inspected	Units Defective	Defect %
Building Center	222669	5079	2,28%
Club Hardware	74992	3071	4,10%
Incity Timber	10935	460	4,21%
Jone's Paint Store	174	3	1,72%
King's Supply Company	77691	909	1,17%
Local Storage Hardware	284	2	0,70%
Pacifica Timber	6412	1163	18,14%
Totals	**393157**	**10687**	**2,72%**

Table 8: Summarized Supplier Data during the trial period with Watson Lumber Company

Supplier	Units Received	% of inspected units
Building Center	273989	85,66%
Club Hardware	91877	88,02%
Incity Timber	12813	83,93%
Jone's Paint Store	213	81,69%
King's Supply Company	87734	83,02%
Local Storage Hardware	291	100,00%
Watson Lumber Company	10395	89,36%
Totals	**477312**	**85,67%**

Table 9: Summarized Quality Data for each supplier during the trial period with Watson Lumber Company

Supplier	Units Inspected	Units Defective	Defect %
Building Center	234689	4979	2,12%
Club Hardware	80867	3202	3,96%
Incity Timber	10754	582	5,41%
Jone's Paint Store	174	1	0,57%
King's Supply Company	72841	692	0,95%
Local Storage Hardware	291	5	1,72%
Watson Lumber Company	9289	524	5,64%

Totals	**408905**	**9985**	**2,44%**

Table 10: Current Summarized Supplier Data (with Highe's Lumber)

Supplier	Units Received	% of inspected units
Building Center	624388	81,94%
Club Hardware	243912	81,00%
Highe's Lumber	20920	82,61%
Incity Timber	23983	78,94%
Jone's Paint Store	666	100,00%
King's Supply Company	236500	79,54%
Local Storage Hardware	764	93,59%
Totals	**1151133**	**35,52%**

Table 11: Current Summarized Quality Data (with Highe's Lumber)

Supplier	Units Inspected	Units Defective	Defect %
Building Center	511612	11310	2,21%
Club Hardware	197564	9734	4,93%
Highe's Lumber	17282	1352	7,82%
Incity Timber	18933	1049	5,54%
Jone's Paint Store	666	7	1,05%
King's Supply Company	188110	2121	1,13%
Local Storage Hardware	715	11	1,54%
Totals	**934882**	**25584**	**2,74%**

Table 12: Material MAT002 Supplier comparison

Supplier	Units Received	Units Inspected	% of inspected units	Units Defective	Defect %
Highe's Lumber	20920	17282	82,61%	1352	7,82%
Pacifica Timber	7719	6412	83,07%	1163	18,14%
Watson Lumber Company	10395	9289	89,36%	524	5,64%

Table 13: Task operation times

Task	Minimum Time (sec)	Average Time (sec)	Maximum Time (sec)	Max units per hour	Avg units per hour	Min units per hour
Assembly	8	10	12	450	360	300
Cure	12	15	18	300	240	200
Cutting	24	30	36	150	120	100
Drilling	16	20	24	225	180	150
Painting	16	20	24	225	180	150
Routing	16	20	24	225	180	150
Sanding	12	15	18	300	240	200
Paint Curing	3600	3600	3600	40	40	0

Buffers

Buffers are currently located in multiple spots around the workshop, so workers have to walk less to supply each task. This might increase the inventory costs as well as the total production times.

Conceptual design

The contract with Highe's Lumber for oak planks will expire in three months, somanagement has tasked you with determining whether the Simio Shelving Shop should renew the contract or accept one of the contracts presented by two potential suppliers,Watson Lumber Company or Pacifica Timber.

The components most important of system are the Metrics that interest management:
- Throughput
- Average unit time in system
- Workstation utilization
- Average inventory cost for each buffer
- Number of stockouts
- Unmet demand by day

The data given by the company needed to be analyzed first, so a statistical distribution was developed for every operation throughout the process, and for every product to be obtained. There are 32 operations and 6 products, leaving a total of 192 statistical distributions. That is the base for the input through the simulated model. A table can be found as an attachment.

Regarding the contract renewal, quality data was analyzed, and the comparison can be observed in **Table 11**.

Also, regarding the challenge problem, where a decision needs to be made about the contract with Highe's Lumber. It is considered that no simulation is required, since the sample size for each supplier is large enough and the difference between each is significant. So assuming similar and competitive prices, this choice could be made merely by comparing these samples, and the one that had less rejections was Watson Lumber. However this rejection was not much smaller than Highe's Lumber. Pacifica, on the other hand, was way out of the picture.

Verification and validation

All servers and sources were made according to the design and process of the company. Also all the times provided were analyzed and their distribution was determined under excel tools. All data or system added was under the document delivered and also that the system does not present any malfunction.

Data Analysis and validation:

Table 14. Data analysis of production of each station and the shelf type.

Operacion	Tiempo promedio en min	Estaciones	unidades diarias
Assemble	6.288720695	3	228.981389
Attach_BP	2.115449835	1	226.9020953
BP_Cut	2.445514326	1	196.2777298
BP_Drill	8.145658915	3	176.7812788
BP_Paint	3.381811263	2	283.8715485
BP_Sand	4.583625192	2	209.4412086
FA_Assemble	1.79158179	1	267.9196689
FA_Cut	4.160505421	2	230.7411968
FA_Drill	7.362989973	3	195.5727232
Paint	7.713979	3	186.6740887
SA_Assemble	3.951259293	2	242.9605168
SA_Cut	3.135249042	1	153.0978859
SA_Drill	9.365275172	3	153.759497
SA_Routing	0.718125	1	668.4073107
Sand	10.37606676	4	185.041215

Each station of the simio Shelf Shop was analyzed to determine their daily production and to establish possible bottle necks.

Table 15. Data analysis of production time of each station.

Promedio de Tiempo de procesamient	Etiquetas de column							
Etiquetas de fila		B22	B23	B24	O22	O23	O24	Total general
Assemble	5.887383177	4.3698701	6.4749579	8.5037598	4.3532209	6.4611111	8.5091995	6.2887207
Attach_BP	2.114948454	1.6269941	2.1528796	2.6751075	1.6293532	2.1510511	2.6996902	2.1154498
BP_Cut	2.44659864	2.1265252	2.618401	2.6242038	2.1255935	2.6313228	2.6229762	2.4455143
BP_Drill	7.98231884	5.6642317	8.3412582	11.022573	5.6271819	8.3589905	10.795184	8.1456589
BP_Paint	3.498484848	2.4610422	3.4787879	4.4686901	2.4710249	3.4504604	4.4359173	3.3818113
BP_Sand	4.676106195	3.267735	4.6727199	6.1878936	3.1920611	4.6470617	6.1875792	4.5836252
FA_Assemble	1.795377129	1.7930122	1.7928634	1.7921791	1.7876233	1.7894394	1.7926932	1.7915818
FA_Cut	4.188235294	3.8364013	4.3388191	4.3509174	3.8387775	4.3301955	4.3539141	4.1605054
FA_Drill	7.323132184	6.0625622	7.4989167	8.8712963	6.0896667	7.3973765	8.8696643	7.36299
Paint	7.529402516	5.4719949	8.0083754	10.484096	5.4531609	7.9582849	10.410142	7.713979
SA_Assemble	3.9	2.2165404	4.074386	5.9488021	2.2352802	4.0910546	5.9575866	3.9512593
SA_Cut	2.909090909	2.0332905	3.2154399	4.3454969	2.0360248	3.2238921	4.3527968	3.135249
SA_Drill	9.547424242	5.6939506	9.6512949	13.638479	5.6558414	9.691	13.685521	9.3652752
SA_Routing	0.748062016	0.5318408	0.7339837	0.9372917	0.5340058	0.7305305	0.9297074	0.718125
Sand	10.55980392	7.1675127	10.684798	14.14065	7.1414401	10.761289	14.221307	10.376067
#¡VALOR!	4.447894737	3.4798246	5.1724859	6.2006098	3.7187225	5.0630464	6.6122372	4.9473974
Total general	**5.062824923**	**3.6221936**	**5.1873304**	**6.5621869**	**3.5922039**	**5.1873607**	**6.590819**	**5.0208241**

Figure 4. Simio run of the actual process.

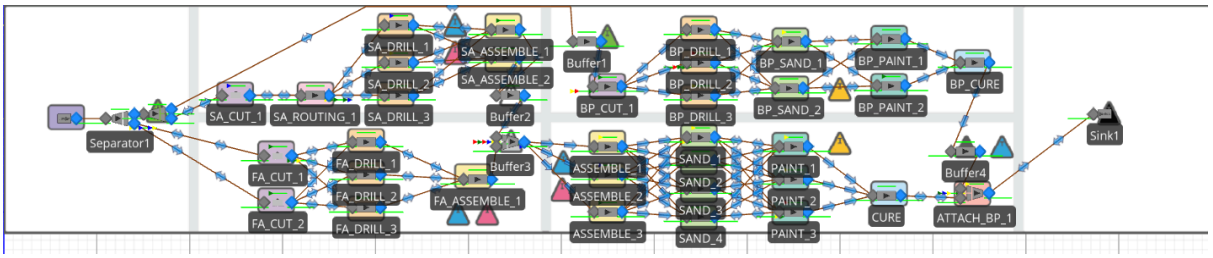
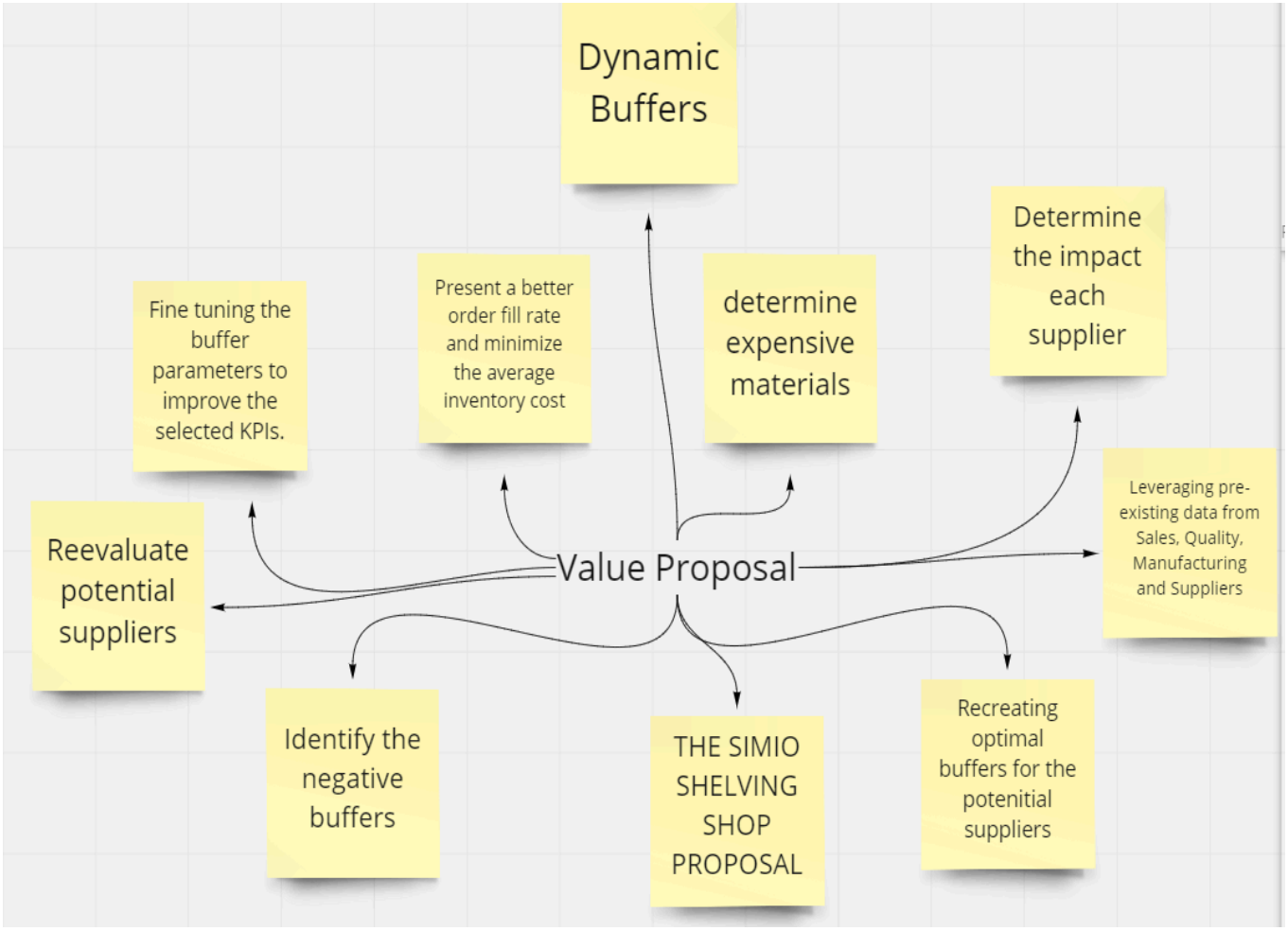


Table 16. Simio Shelf Shop Assembly simulation on a 6 week run.

Sink	Sink1	InputBuffer	Throughput	NumberEntered	Total	5,359.0000
				NumberExited	Total	5,359.0000
		[DestroyedEntities]	FlowTime	TimeInSystem	Observations	5,359.0000
					Minimum (Hou...	0.0000
					Maximum (Ho...	0.0000
					Average (Hou...	0.0000

Value Proposal

The Simio Shelving Shops has multiple areas that can be improved with the Dynamic Buffers method, creating a better order fill rate and minimizing the average inventory cost. It is important to analyze every area and their points of improvement to determine if it is convenient for the company to establish every proposal.



Determine the buffers locations: Establishing the correct location of each buffer will allow the company to have a better control on the possible improvement points.

Fine tuning the buffer parameters to improve the selected KPIS: The buffer will need the correct parameters so it can prevent failure in the manufacturing process. The simulation will establish the data to determine those parameters.

Reevaluate Potential suppliers and determine the ones that fulfill the company quality parameters: Suppliers are an essential part of the manufacturing process that need to always be on time with their orders and offer the best quality/price ratio.

Recreate the manufacturing process and space: Recreating the space of manufacturing will allow us to identify better ways of assembling each shelf type at the lowest cost possible.

Each idea will be simulated in simio separately so we can analyze the impact individually in the process. Randomness of situations, supplies runouts, and labor dysfunction will be useful to evaluate in the simulation.

Alternatives Evaluation

Alternative 1: Eliminating Bottlenecks.

Adding an extra station in the bottle neck will improve its throughput and eliminate the bottle neck.

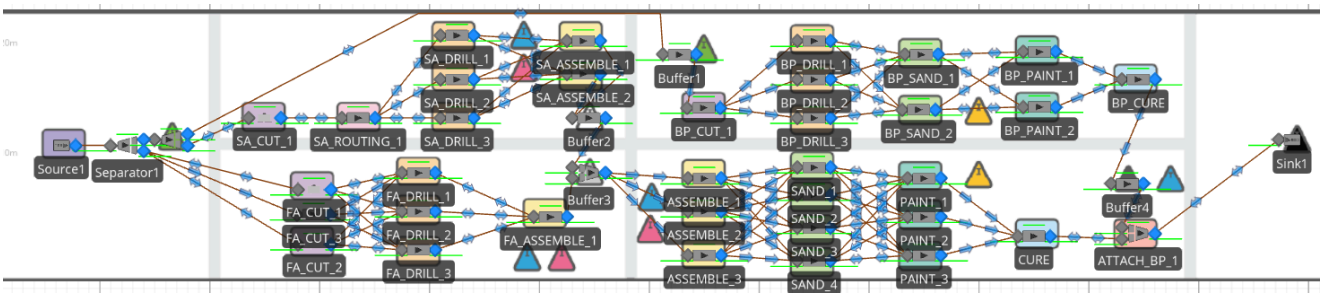


Table 17. Alternative 1 run results

Object Type	Object Name	Data Source	Category	Data Item	Statistic	Average Total
Server	FA_CUT_3	[Resource]	ResourceState	TimeStarved	Average (Hou...	1,008.0000
					Occurrences	1.0000
					Percent	100.0000
					Total (Hours)	1,008.0000
		OutputBuffer	Content	NumberInStation	Average	11.2283
					Maximum	25.0000
					Minimum	0.0000
			HoldingTime	TimeInStation	Average (Hou...	4.7106
					Maximum (Ho...	63.9302
					Minimum (Hou...	0.0000
			Throughput	NumberExited	Total	2,244.0000
				NumberEntered	Total	2,256.0000
		Processing	Content	NumberInStation	Average	0.2781
					Maximum	1.0000
					Minimum	0.0000
			EntryQueue	TimeWaiting	Average (Hou...	0.7213
					Maximum (Ho...	2.9668
					Minimum (Hou...	0.0000
				NumberWaiting	Average	1.6143
					Maximum	24.0000
					Minimum	0.0000
			HoldingTime	TimeInStation	Average (Hou...	0.1242
					Maximum (Ho...	0.8583
					Minimum (Hou...	0.0547
			Throughput	NumberExited	Total	2,256.0000
				NumberEntered	Total	2,256.0000
Sink	Sink1	[DestroyedEntities]	FlowTime	TimeInSystem	Average (Hou...	0.0000
					Maximum (Ho...	0.0000
					Minimum (Hou...	0.0000
					Observations	6,172.0000
		InputBuffer	Throughput	NumberExited	Total	6,172.0000
				NumberEntered	Total	6,172.0000

Alternative 2: Eliminate Extra Work.

All the personal that is at a station that becomes a bottleneck of the whole process, will not be responsible of transporting the material just so they can focus on working in their station and save time

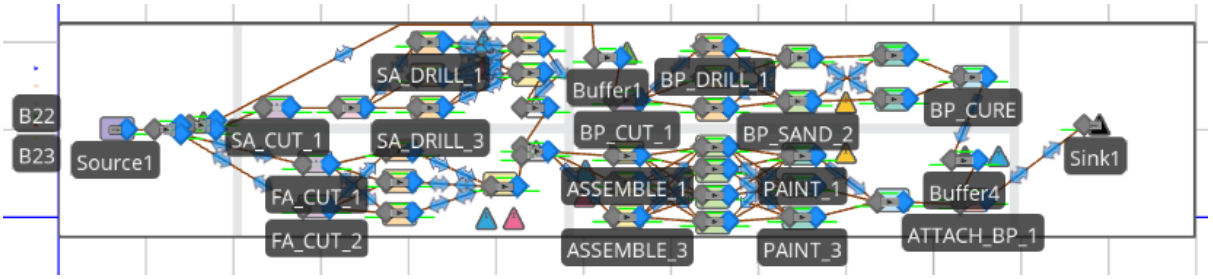


Table 18. Alternative 2 run results

Object Type	Object Name	Data Source	Category	Data Item	Statistic	Average Total
Sink	Sink1	[DestroyedEntities]	FlowTime	TimeInSystem	Average (Hou...	0.0000
					Maximum (Ho...	0.0000
					Minimum (Hou...	0.0000
					Observations	5,471.0000
		InputBuffer	Throughput	NumberExited	Total	5,471.0000
				NumberEntered	Total	5,471.0000

Alternative 3 Alternating Working Hours.

Propose to the workers of the bottleneck to arrive early to the station that have production delays and offer them to live 2 hours earlier, in that way the station can eliminate the delay before the production of the shop starts and improve the throughput of the whole factory. For every worker that is offered to arrive and leave early, one from the quicker stations is offered to arrive 2 hours later, and leave 1 hour later. They swap stations during lunch time to extend the overall time the bottleneck station operates, and sacrificing up to 4 hours in the quicker stations.

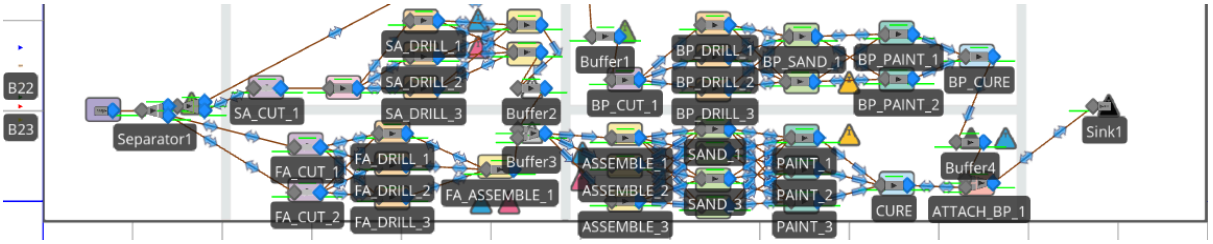


Table 19. Alternative 3 run results

Object Type	Object Name	Data Source	Category	Data Item	Statistic	Average Total
Sink	Sink1	[DestroyedEntities]	FlowTime	TimeInSystem	Average (Hou...	0.0000
					Maximum (Ho...	0.0000
					Minimum (Hou...	0.0000
					Observations	5,765.0000
		InputBuffer	Throughput	NumberEntered	Total	5,765.0000
				NumberExited	Total	5,765.0000

Alternative comparison and final recommendations

The three alternatives can be implemented at the same time in the process, but this can mean an upper production cost that can be higher than the throughput it produces. To minimize costs the best alternative is the number 2 and 3 where the only cost is time, but that time is used to eliminate bottlenecks and improve the throughput.

Table 20. Weekly production by alternative

Alternative	Weekly shelves production
Actual	893
1	1028
2	911
3	960

Implementation guide and final recommendations.

According to the simulation results all alternatives improve the production in comparison with the actual run, but as mentioned before only alternatives 2 and 3 represent a time only cost, the alternative 1 requires investment in capacitation and personal. To obtain a similar or better result as alternative 1, alternative 2 and 3 can be added to the process and establish a better throughput without adding more personal . These 2 alternatives represent a dynamic buffer that can eliminate delays in any station because it does not require capacitation or special tools to function. For a final recommendation the principal material provider should be Watson Lumber Company, as analyzed in tables 8, 9 and 10, they have the best material quality even though they take time to deliver. It is better to prepare with anticipated time the supplies and receive them with high quality than to receive in a short time and losing some supplies because of quality controls.